Jeongmin Bae

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Professional Summary

I am a Postdoctoral Research Fellow at the University of Melbourne and received my Ph.D. in Artificial Intelligence from Yonsei University in 2026. My research focuses on 3D/4D computer vision, neural rendering, and dynamic scene reconstruction, with particular expertise in Gaussian Splatting and Neural Radiance Field (NeRF)-based methods for dynamic and egocentric scenarios.

I develop learning-based approaches for reconstructing and modeling complex dynamic environments from visual observations, with a focus on geometry, appearance, and motion over time. My work has been published at leading computer vision and graphics venues, including ECCV, ICCV, AAAI, and SIGGRAPH.

My current research interests include real-time 3D perception, hand-object interaction, and scene and motion understanding. I am interested in developing 3D/4D representations that work well in less controlled real-world settings and can be extended with foundation-model features for scene, action, and human-object interaction understanding.

Research Interests

3D/4D Computer Vision, Neural Rendering, Dynamic Scene Reconstruction, Hand-Object Interaction, Egocentric Vision, Spatial Computing, Generative AI

Technical Skills

3D Vision & Spatial Computing: Gaussian Splatting, Neural Radiance Fields (NeRF), Dynamic Scene Reconstruction, Camera Calibration, LiDAR Processing, Multi-Camera Systems

Machine Learning: PyTorch, Foundation Models, Generative Models

Programming & Tools: Python, Git, Docker, Linux

Publications

- **Hand-4DGS: Feed-Forward 3D Gaussian Splatting for 4D Hand Reconstruction from Egocentric Videos** — *arXiv 2026*
  __Jeongmin Bae__, Seoha Kim*, Marc Pollefeys, Madhi Rad, Youngjung Uh†, Taein Kwon†*
  - This work proposes a generalizable feed-forward Gaussian splatting model that enables hand reconstruction in egocentric videos where existing methods often struggle. To this end, we utilize hand template priors and learned representations from vision foundation models.
- **4D Scaffold Gaussian Splatting with Dynamic-Aware Anchor Growing for Efficient and High-Fidelity Dynamic Scene Reconstruction** — *AAAI 2026*
  Woong Oh Cho, In Cho, Seoha Kim, __Jeongmin Bae__, Youngjung Uh, Seon Joo Kim
  - This paper introduces a 4D anchor-based framework for dynamic scene reconstruction that reduces storage costs while maintaining the high visual quality and fast rendering speed of 4D Gaussian methods.
- **Compensating Spatiotemporally Inconsistent Observations for Online Dynamic 3D Gaussian Splatting** — *SIGGRAPH 2025*
  Youngsik Yun, __Jeongmin Bae__, Hyunseung Son, Seoha Kim, Hahyun Lee, Gun Bang, Youngjung Uh
  - This paper shows that online dynamic scene reconstruction via 3D Gaussians can suffer from temporally inconsistent reconstruction, especially in static regions, depending on the noise in the observations, and addresses this by introducing a residual map.
- **Rethinking Open-Vocabulary Segmentation of Radiance Fields in 3D Space** — *AAAI 2025*
  Hyunjee Lee, Youngsik Yun*, __Jeongmin Bae__, Seoha Kim, Youngjung Uh*
  - This paper aims to perform clear open-vocabulary semantic segmentation in 3D space and proposes an evaluation protocol that assesses 3D geometry and segmentation simultaneously.
- **Per-Gaussian Embedding-Based Deformation for Deformable 3D Gaussian Splatting** — *ECCV 2024*
  __Jeongmin Bae__, Seoha Kim*, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper demonstrates that existing deformable 3D Gaussian Splatting models fail to reconstruct complex dynamic scenes and addresses this by replacing the input of the deformation function with learnable embeddings.
- **Optimizing Dynamic NeRF and 3DGS with No Video Synchronization** — *ECCV 2024 Wild3D workshop*
  Seoha Kim, __Jeongmin Bae__*, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper is an extension of Sync-NeRF (Seoha Kim et al., 2024) with the addition of dynamic 3D Gaussian Splatting.

- **Sync-NeRF: Generalizing Dynamic NeRFs to Unsynchronized Videos** *AAAI 2024*
  Seoha Kim, **Jeongmin Bae***, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper demonstrates the failure of existing models in dynamic scene reconstruction from unsynchronized multi-view videos and proposes a solution by introducing per-camera time offsets to model calibrated time-dependent representations.
- **BallGAN: 3D-aware Image Synthesis with a Spherical Background** *ICCV 2023*
  *Minjung Shin, Yunji Seo, **Jeongmin Bae**, Young Sun Choi, Hyunsu Kim, Hyeran Byun, Youngjung Uh*
  - This paper aims to learn the 3D geometry of the foreground from a large image collection by modeling the background as a 2D sphere.
- **FurryGAN: High Quality Foreground-aware Image Synthesis** *ECCV 2022*
  ***Jeongmin Bae**, Mingi Kwon, Youngjung Uh*
  - This paper aims to train a GAN to automatically separate fine-detailed foregrounds (such as fur) from large collections of single-view images without using external models or additional supervision.

EXPERIENCE

- **University of Melbourne – School of Computing and Information Systems [🏠]** *Mar. 2026 – Feb. 2027*
  Postdoctoral Research Fellow
  - Postdoctoral Research Fellow in the Faculty of Engineering and Information Technology (FEIT), supervised by Prof. Taehyun Rhee.
  - Collaborating on dynamic scene reconstruction and mixed reality projects using multiple 360° camera–LiDAR sensor pairs.
- **Electronics and Telecommunications Research Institute (ETRI) [🏠]** *Nov. 2022 - Mar. 2025*
  Academic-Research Cooperation
  - Research on enhancing rendering quality in deformable 3D Gaussian Splatting for complex dynamic scenes.
  - Research on dynamic scene reconstruction from unsynchronized video datasets using neural radiance fields.
  - Comparison of implicit neural representation models for dynamic scene reconstruction and model reproduction (e.g. DyNeRF, CVPR 2022).

ACADEMIC SERVICE

Peer Reviewer: ECCV, ICCV, CVPR, NeurIPS, IEEE TVCG, IEEE TPAMI

EDUCATION

- **Yonsei University - Visual Intelligence Lab[🏠]** *Feb. 2022 - Feb. 2026*
  Ph.D. in Artificial Intelligence (Completed), advised by Prof. Youngjung Uh Seoul, South Korea
  - GPA: 4.14/4.3
- **Kookmin University** *Feb. 2013 - Feb. 2021*
  B.S., School of Electronical Engineering Seoul, South Korea
  - Grade: 3.95/4.5 (Major GPA: 4.06/4.5)

PATENTS

[2025] METHOD AND APPARATUS FOR DECODING IMAGE,
KR 10-2024-0043684

[2023] Method and apparatus for representing dynamic neural radiance fields from unsynchronized videos,
KR 10-2023-0105173

AWARDS

- **Excellence Award** *2019*
  Python Programming Using Big Data / Samsung Multicampus
  - Received an exceptional excellence award for actively helping and collaborating with others during course completion.
- **Departmental Top Student Scholarship** *2018*
  Kookmin University
  - Won a scholarship in the first semester of my third year as an electronical engineering student, ranked number one out of 222 students in the entire department for the semester.

Jeongmin Bae - Portfolio

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Overview

My research focuses on spatial representation learning from real-world visual data, including dynamic 3D/4D reconstruction, Gaussian Splatting, NeRF-based scene modeling, egocentric hand reconstruction, and robust reconstruction from imperfect observations.

I am currently interested in developing 3D/4D reconstruction and scene understanding methods that work well in less controlled real-world settings, where visual observations often include challenges such as incomplete data, noise, occlusions, unsynchronized videos, and moving cameras.

At the University of Melbourne, I am currently a Postdoctoral Research Fellow leading a mixed-reality project on real-world scene reconstruction using multiple 360° camera–LiDAR sensor pairs. I received my Ph.D. in Artificial Intelligence from Yonsei University in February 2026.

Ongoing Research Direction

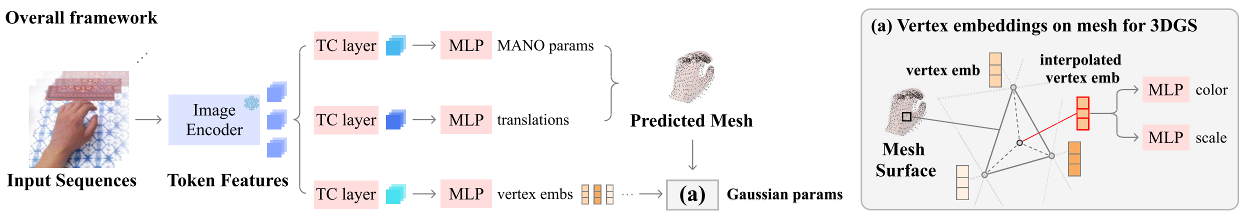
Hand-4DGS: Feed-Forward 3D Gaussian Splatting for 4D Hand Reconstruction from Egocentric Videos

Ongoing work, arXiv 2026, Co-first author

Problem Definition

- Reconstructing 4D hands from egocentric videos is important for applications such as AR/VR, AI glasses, robot manipulation, and teleoperation.
- Egocentric hand reconstruction is challenging due to single-view depth ambiguity, rapid hand motion, severe occlusion, and moving cameras.
- Existing methods often rely on multiple viewpoints, which limits their applicability to wearable and egocentric settings. Per-video optimization methods are also too slow for real-time applications.
- This motivates a feed-forward framework that can rapidly reconstruct 4D hands from single-view egocentric videos and generalize to unseen videos.

Approach



- I introduced a mesh-guided 3D hand representation that uses the MANO hand mesh as a structural prior and assigns learnable embeddings to mesh vertices.
- The method samples Gaussians on the mesh surface and interpolates vertex embeddings to define the geometry and appearance of each Gaussian.
- It uses vision foundation model features together with temporal convolution layers to aggregate information from adjacent frames and reduce temporal jitter.
- Outputs from an existing hand pose estimation model are used as initial supervision, while image-level reconstruction losses help refine the hand appearance and pose during training.
- This feed-forward pipeline enables instant 4D hand reconstruction from RGB video inputs without per-video optimization.

Results

- Implemented a feed-forward hand reconstruction pipeline capable of real-time inference and more stable reconstruction than per-scene optimization baselines. (Inference speed: ∼60 FPS; EVA PSNR: 23.56 → Ours: 24.66.)
- Demonstrated instant 4D hand reconstruction from RGB inputs on unseen videos.
- Reduced inference-time dependency on pose estimators while improving hand pose and appearance reconstruction through image-level supervision.

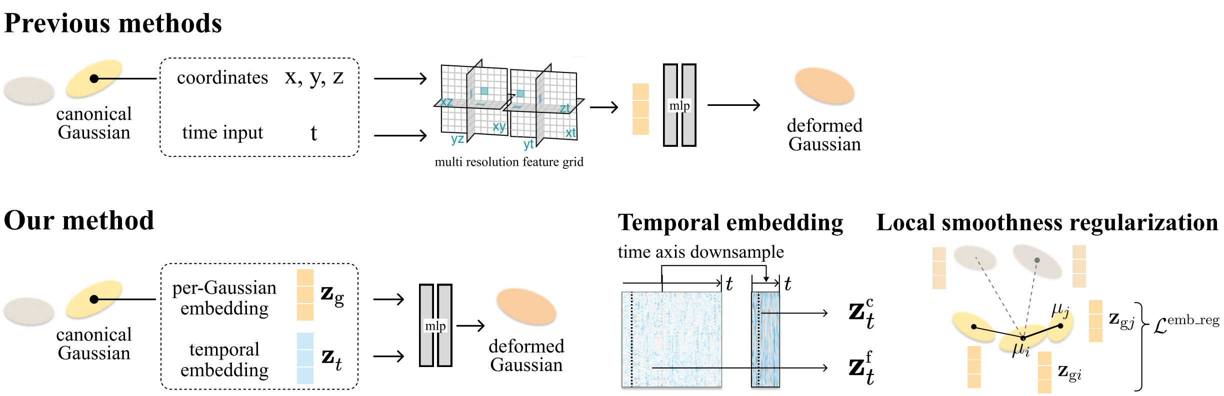
E-D3DGS: Per-Gaussian Embedding-Based Deformation for Dynamic 3D Gaussian Splatting

ECCV 2024, Co-first author

Problem Definition

- Existing deformable 3D Gaussian Splatting methods often produce blurry results or fail to accurately represent dynamic regions in complex scenes.
- Many methods predict temporal changes from the coordinates of Gaussian primitives in canonical space, which can make it difficult to model different motions for nearby scene elements.
- As scene dynamics become more complex, these approaches become increasingly dependent on the capacity of MLPs or the resolution of feature grids, increasing model complexity and memory cost.
- This motivates a dynamic scene representation that can effectively model complex motion without relying heavily on deep networks or high-resolution grid representations.

Approach



- I proposed an embedding-based dynamic scene representation that assigns an individual learnable latent embedding to each Gaussian primitive.
- A per-Gaussian latent embedding captures Gaussian-specific deformation, while a global temporal embedding represents the scene state at each time step.
- The deformation of each Gaussian is predicted by a shallow decoder using its per-Gaussian embedding and the temporal embedding for the corresponding time step.
- I used a coarse-fine deformation strategy to separate large-scale motion from fine-grained local details by dividing temporal embeddings into coarse and fine embeddings.
- I also introduced local smoothness regularization based on the assumption that nearby Gaussians tend to undergo similar deformations, helping preserve texture details in dynamic regions.

Results

- On the Neural 3D Video Dataset, the proposed method improved reconstruction quality, rendering speed, and model efficiency compared to 4DGaussians. (PSNR: 30.71 $\rightarrow$ 31.31; Rendering speed: 51.9 FPS $\rightarrow$ 74.5 FPS; Model size: 59 MB $\rightarrow$ 35 MB.)
- The method also achieved better results than baselines in monocular video settings, demonstrating robust performance across different capture setups.
- This work was published at ECCV 2024, and related patent applications were filed in Korea and the United States (KR 10-2024-0043684 / US 2025/0308078 A1).

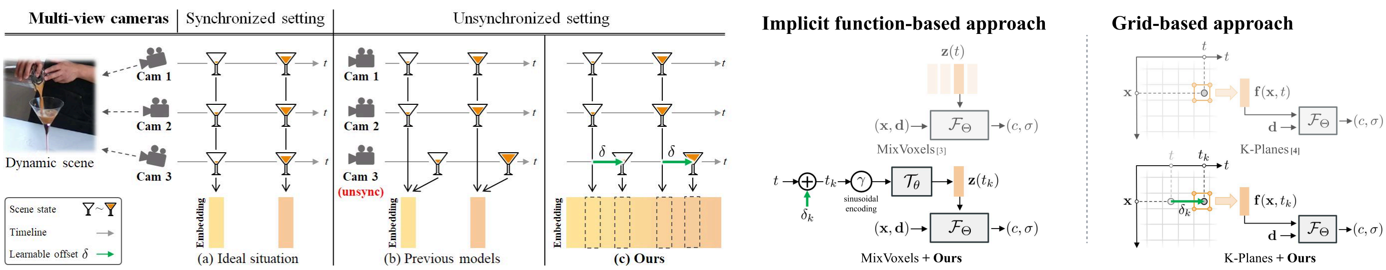
Sync-NeRF: Generalizing Dynamic NeRFs to Unsynchronized Videos

AAAI 2024, Co-first author

Problem Definition

- Existing multi-view dynamic reconstruction methods often assume that images with the same frame index capture the same moment in time.
- In practice, capture timing can differ across cameras. In a commonly used multi-view dynamic video dataset, some views were significantly misaligned despite being captured with synchronization equipment such as timecode systems.
- Training with unsynchronized views can cause ghosting and artifacts in dynamic regions, which degrades reconstruction quality.
- This motivates a dynamic scene reconstruction method that can handle multi-view capture data with timing misalignment.

Approach



- I co-proposed a training strategy that directly compensates for camera-specific time offsets instead of discarding unsynchronized multi-view videos.
- The method introduces a learnable time offset for each camera and optimizes it jointly with the dynamic NeRF representation.
- This allows the model to align observations from different cameras while learning a time-dependent 3D scene representation.
- The strategy is compatible with both implicit-function-based temporal embeddings and grid-based temporal representations.

Results

- The proposed method improved reconstruction quality across multiple baselines under unsynchronized multi-view settings. (MixVoxels: PSNR 29.96 → 30.53; K-Planes: PSNR 29.16 → 30.44.)
- The learned time offsets compensated for capture-time misalignment, enabling robust dynamic reconstruction from multi-view videos with timing errors.
- This work was published at AAAI 2024, and related patent applications were filed in Korea and the United States (KR 10-2023-0105173 / US 2025/0054224 A1).

Other Publications

- **4D Scaffold Gaussian Splatting with Dynamic-Aware Anchor Growing for Efficient and High-Fidelity Dynamic Scene Reconstruction**, AAAI 2026. Introduces a 4D anchor-based framework that reduces storage cost while maintaining high visual quality and fast rendering speed for dynamic scene reconstruction.
- **Compensating Spatiotemporally Inconsistent Observations for Online Dynamic 3D Gaussian Splatting**, SIGGRAPH 2025. Addresses temporally inconsistent reconstruction in online dynamic 3DGS by introducing a residual map.
- **Rethinking Open-Vocabulary Segmentation of Radiance Fields in 3D Space**, AAAI 2025. Proposes an evaluation protocol for open-vocabulary 3D segmentation that jointly assesses geometry and segmentation quality.
- **BallGAN: 3D-aware Image Synthesis with a Spherical Background**, ICCV 2023. Learns foreground 3D geometry from image collections by modeling the background as a 2D sphere.
- **FurryGAN: High Quality Foreground-aware Image Synthesis**, ECCV 2022. Trains a GAN to separate fine-detailed foregrounds from single-view image collections without external models or additional supervision.